CS 595 - Medical Informatics & AI

Project Proposal

DischargeIQ

Multi-Agent Post-Discharge Diagnostic Intelligence System

Connecting Diagnosis, Medications, and Recovery in Plain Language at the Point of Discharge

1. Project Definition

Project ID / Title	CS595-2026-DischargeIQ Multi-Agent Post-Discharge Diagnostic Intelligence System
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LOF Pillar	Patient Engagement
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2. Introduction

2a. Problem Description - NABC Analysis

NABC Framework - DischargeIQ	
Need	• The hospital discharge moment is the most dangerous handoff in American healthcare. Every year, approximately 36 million patients are discharged from US hospitals - and a majority leave without fully understanding what happened to their body. • Although 95.6% of patients report understanding the reason they were hospitalized, only 59.6% can accurately describe their diagnosis when interviewed after discharge. The gap between perceived and actual understanding is invisible to the care team because nobody checks. • Studies show 40-80% of patients forget or misunderstand what was discussed during their hospital stay. Patients recall less than 15% of verbal discharge instructions. Over 80% of Epic discharge documents exceed the AMA-recommended 6th grade reading level, yet 36% of American adults have basic or below-basic health literacy.

	• The diagnosis problem is deeper than document readability. Research shows that almost 40% of initial pneumonia diagnoses are later revised, and a 2024 JAMA study found 23% of patients transferred to an ICU or who died had a missed or delayed diagnosis. Patients are discharged with diagnostic uncertainty baked into the document - and have no way to recognize or act on it. • Medication misunderstanding compounds the diagnosis gap. Error rates are highest for medication side effect comprehension (50%), adherence to medication dose (34%), and return precaution instructions (78%). Patients who make comprehension errors are nearly 9 times more likely to make adherence errors. • The financial stakes are institutional: $41.3 billion annually in 30-day readmission costs, with 27% estimated to be preventable. CMS penalizes hospitals up to 3% of their Medicare reimbursement under the Hospital Readmissions Reduction Program for excess readmissions - creating a direct financial incentive for hospitals to invest in discharge comprehension solutions. • There is no existing patient-facing tool that connects a patient's specific diagnosis to their specific medications to their specific recovery trajectory in a single plain-language, clinician-validated output.
Approach	• A patient is discharged from the hospital. They open DischargeIQ on any mobile device and upload a PDF of their discharge document. Five specialized AI agents run sequentially, each handling a distinct task type that justifies the multi-agent architecture. • Agent 1 - Extraction Agent: Parses the discharge document and identifies all structured information: primary and secondary diagnoses, procedures performed, new and changed medications, discontinued medications, activity and dietary restrictions, follow-up appointment details, and listed red flag symptoms. • Agent 2 - Diagnosis Explanation Agent: Takes the primary diagnosis and generates a plain-language explanation of what the condition is, why the patient developed it, what the treatment targeted, and what the expected recovery trajectory looks like. For the MVP, templates are validated against published clinical guidelines for 5 diagnoses: heart failure, COPD, diabetes management, hip replacement, and one surgical case. • Agent 3 - Medication Rationale Agent: For each medication, connects the drug to the diagnosis and explains why that specific medication addresses that specific condition, what the patient will feel as it works, what side effects are expected versus which require a call or ER visit, and what happens if doses are missed. • Agent 4 - Recovery Trajectory Agent: Generates a condition-specific recovery timeline telling the patient what the first week looks like, what functional limitations are normal versus alarming, what symptoms indicate progress versus deterioration, and when they should expect to feel meaningfully better. • Agent 5 - Alert and Escalation Agent: Extracts every red flag symptom from the document, supplements with condition-specific flags not always listed on discharge documents, and produces a clear decision tree: call your doctor, go to the ER immediately, or normal and expected. • Output: Structured, mobile-friendly page at a 6th grade reading level with five clearly labeled sections. Tech stack: FastAPI backend, Claude API for all five agents, Streamlit or React frontend, Neon PostgreSQL for patient history. Optional SMS daily checklist is a stretch goal.
Benefits	• Patients leave the hospital with something they can actually use. The 40-80% who currently forget or misunderstand critical information have a persistent, accessible reference they can return to at 2am when they are anxious and cannot reach their provider.

	• The diagnosis-to-medication connection is the key innovation no existing tool makes. When a patient understands why they are on Furosemide because of fluid retention caused by their heart failure - not just that they were prescribed a diuretic - their adherence behavior changes qualitatively. • The recovery trajectory gives patients a framework to interpret what they are experiencing at home, reducing unnecessary ER visits for normal recovery symptoms and accelerating calls for genuinely alarming ones. • Hospitals gain a measurable, documentable discharge quality improvement intervention directly relevant to CMS readmission penalties. The evaluation framework generates comprehension scores and 30-day outcome data that quality improvement departments need. • For LOF: DischargeIQ is the upstream layer that feeds TheraCareAI. A patient who enters adherence tracking with a clear understanding of their diagnosis and medications produces cleaner behavioral data and more accurate state transitions. DischargeIQ is Box 1, TheraCareAI is Box 2, and together they cover the full patient journey from discharge to sustained adherence.
Competition	• Epic AfterVisit Summary: Auto-generated, written in clinical language, consistently exceeds recommended reading levels (confirmed December 2024 study). No diagnosis explanation layer. No medication rationale. No recovery trajectory. Stops at document delivery, not patient comprehension. • Abstractive Health: Generates better clinical summaries for physicians. Their output stops at the provider desk. Patient-facing explanations are explicitly outside their product scope. Their tool and DischargeIQ are complementary, not competing. • Discharge nurse education: Time-constrained, verbal, not retained. Studies confirm patients recall less than 15% of verbal instructions regardless of education effort. • 1mg, Healthline, WebMD: General condition information not connected to the patient's specific discharge document, specific medications, or specific recovery context. No document parsing. No personalization. • KEY DIFFERENTIATION: Every competitor handles either the document or generic health education. DischargeIQ is the only tool that connects a patient's specific discharge document to a specific diagnosis explanation, to a specific medication rationale, to a specific recovery trajectory, in a single clinician-validated mobile pipeline. That integration - across all five dimensions simultaneously - is the product.

2b. Purpose

DischargeIQ enables patients to photograph or upload their hospital discharge document and immediately receive a clinician-validated, plain-language explanation of their diagnosis, the rationale behind every medication they were prescribed, a condition-specific recovery timeline, and a clear escalation guide for warning symptoms - reducing preventable readmissions by closing the comprehension gap at the most critical handoff in the care journey.

Goals and objectives: Build a working five-agent pipeline within 6 weeks that demonstrates discharge document extraction, AI-based diagnosis explanation, medication rationale generation, recovery trajectory modeling, and alert escalation logic. Produce a deployable MVP with a clinician-validated explanation library covering the 5 highest-readmission diagnoses, a documented evaluation framework measuring pre/post comprehension and reading level, and a pilot pathway for deployment with small scale hospital network.

2c. Scope

Included in Scope

Deliverable	Description
Discharge document upload and extraction	PDF upload (primary for MVP) and photo upload (stretch goal) via mobile-friendly web interface; extraction agent parses diagnoses, medications, restrictions, follow-up dates, and red flags
Five-agent AI pipeline	Extraction, Diagnosis Explanation, Medication Rationale, Recovery Trajectory, and Alert/Escalation agents running sequentially via FastAPI; explicit error handling at each handoff so partial extractions produce partial output, not silent failures
Clinician-validated explanation templates	5 diagnoses for MVP: heart failure, COPD, diabetes management, hip replacement, and one surgical case; each template validated against published clinical guidelines and reviewed within Dr. David Liebovitz's clinical informatics ecosystem before deployment
Plain-language output at 6th grade reading level	Full structured patient-facing output with five clearly labeled sections; Flesch-Kincaid scoring built into pipeline from Sprint 1 and applied to every agent output
Mobile-responsive web UI	Streamlit or React frontend optimized for smartphone use; no app download required
Patient history storage	Lightweight discharge history log in Neon PostgreSQL; document hash and structured extracted fields stored per returning user; full document text not stored
Evaluation framework	Pre/post comprehension scoring, FK reading level delta, medication explanation accuracy, LLM-as-Judge rubric for 20 test cases; Claude API costs estimated before Sprint 2 begins
SMS daily checklist (stretch goal)	Optional two-week daily reminder generated from the recovery trajectory agent; assigned to a secondary owner after core pipeline is stable

Stretch Goals

The following are not guaranteed MVP deliverables but are realistic targets given a 5-person team. Each is assigned to a secondary owner and picked up only after core pipeline is complete and stable.

- Photo and camera upload with OCR: extend the extraction agent to handle photographed discharge documents. Deferred from MVP per professor review; PDF-first is the MVP input format.
- Basic prescription reminder alerts: lightweight in-app or SMS reminder for the first 7 days post-discharge. Natural handoff signal to TheraCareAI for longitudinal tracking.
- Read-only physician summary view: a single-page provider-facing output summarizing what explanation the patient received. Read-only, no EHR write-back.
- Diagnosis templates beyond the core 5: expand after the core pipeline is stable and evaluated.

Excluded from Scope

- Clinical decision support, differential diagnosis generation, or treatment recommendations: crosses a regulatory and liability line. The system must remain in patient education and explanation, not clinical advice. This framing protects the project legally and keeps it deployable without FDA oversight.
- Integration with live EHR systems (Epic, Cerner, ABDM, OpenMRS): requires hospital IT credentials, security reviews, HL7 sandbox access, and NDAs not obtainable within a 6-week academic timeline.
- HIPAA-compliant infrastructure hardening: pre-deployment requirement, not an MVP deliverable. The prototype operates exclusively on de-identified and synthetic discharge documents. Compliance architecture is documented as a Phase 2 requirement.

- Longitudinal adherence tracking: intentionally deferred to TheraCareAI. DischargeIQ outputs are designed as structured handoff data for TheraCareAI's behavioral pipeline.

Future Work

- Non-English language support: the Claude API handles multilingual output natively, and the explanation agent architecture is language-agnostic by design. Hindi, Spanish, and other major language outputs are a Phase 2 addition requiring clinician review of translated templates, not a technical rebuild.
- TheraCareAI API integration: formal handoff endpoint so DischargeIQ outputs feed directly into TheraCareAI's patient state model as structured baseline comprehension data.
- HIPAA-compliant deployment infrastructure: full compliance hardening for real patient data, required before any clinical pilot with actual discharge documents.

2d. Audience

Primary Users

Patients and caregivers in the US who have received a hospital discharge document and need to understand and act on it independently at home. Key characteristics:

- Recently discharged from an acute care hospital after treatment for a high-readmission condition (heart failure, pneumonia, COPD, orthopedic surgery, diabetes management, etc.)
- Access to a smartphone with a camera and mobile internet connection
- Varying health literacy levels; the system is designed for a 6th grade reading level to serve the broadest possible population
- May be experiencing anxiety, pain, or cognitive overload at the point of discharge - the system must be simple enough to use under those conditions
- May have a family caregiver using the tool on behalf of the patient, particularly for elderly or post-surgical patients

Secondary Audiences

- Hospital quality improvement directors seeking a measurable discharge intervention against CMS readmission penalties
- Healthcare researchers and clinicians evaluating AI-assisted patient education tools - specifically Dr. David Liebovitz's clinical informatics ecosystem, which will provide the validation framework for explanation templates
- CS595 course instructors and industry partners assessing a real-world multi-agent pipeline on a high-stakes health equity problem

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2e. System Overview

DischargeIQ is a five-agent AI system built on a FastAPI backend, a mobile-responsive Streamlit or React frontend, and the Claude API as the LLM backbone for all agent tasks. The multi-agent architecture is justified because each stage of the pipeline requires a fundamentally different task type - making a single monolithic prompt a less reliable and less debuggable approach than a sequential agent chain.

Agent	Task Type	Input	Output
1 - Extraction	Document parsing / structured extraction	Raw discharge document PDF	Structured fields: diagnoses, medications, restrictions, follow-up dates, red flags

2 - Diagnosis Explanation	Knowledge retrieval + plain-language generation	Primary and secondary diagnoses	Plain-language explanation of condition, cause, treatment rationale, and expected trajectory
3 - Medication Rationale	Drug-condition linking + patient-safe explanation	Medication list + diagnosis context	Per-medication explanation: purpose, expected effects, side effects, dose-miss consequences
4 - Recovery Trajectory	Condition-specific timeline generation	Diagnosis + procedure performed	Week-by-week recovery guide with normal vs. alarming symptom distinctions
5 - Alert and Escalation	Risk extraction + decision tree generation	Red flags from document + condition context	Clear escalation guide: call doctor / go to ER / normal - with specific symptom triggers

Tech Stack Component	Technology
Backend / Orchestration	FastAPI (Python) - manages sequential agent execution, handles errors, passes outputs between agents
LLM / Agent Engine	Claude API (Anthropic) - powers all five agents with specialized system prompts per agent
Frontend	Streamlit or React - mobile-responsive, no app download required
Database	Neon PostgreSQL - stores patient discharge history and past outputs
Document Input	Photo upload (JPEG/PNG) or PDF upload; extraction agent handles format normalization
Evaluation	Flesch-Kincaid readability scoring, LLM-as-Judge rubric (adapted from Dr. Liebovitz's framework), manual clinician review of explanation templates
Infrastructure Base	Dr. David Liebovitz's GitHub repository (FastAPI + Redis + Neon PostgreSQL + Streamlit stack)

2f. Glossary / Terminology

Term	Definition
DischargeIQ	The project name for the multi-agent post-discharge intelligence system described in this proposal
Multi-agent system	An AI architecture where distinct AI agents each perform a specific task type, passing structured outputs between them via an orchestration layer
HRRP	Hospital Readmissions Reduction Program - CMS program that penalizes hospitals with excess 30-day readmission rates by reducing Medicare reimbursements by up to 3%
Health literacy	A patient's ability to obtain, process, and understand basic health information to make appropriate health decisions
Flesch-Kincaid	A validated readability scoring system that estimates the US grade level required to understand a piece of text; AMA recommends patient materials at grade 6 or below

LLM-as-Judge	An evaluation methodology where a language model scores generated outputs against a structured rubric; adapted from Dr. Liebovitz's clinical AI evaluation framework
Recovery Trajectory	A condition-specific timeline describing expected functional progression, normal symptoms, alarming symptoms, and milestone checkpoints after hospital discharge
Medication Rationale	A patient-facing explanation connecting a prescribed drug to the specific condition it addresses, including expected effects, side effects, and consequences of non-adherence
Orchestration layer	The FastAPI backend component that coordinates sequential agent execution, passes outputs between agents, handles errors (e.g., unrecognized diagnoses), and manages retry logic
NABC	Need, Approach, Benefits, Competition - the project evaluation framework used in CS595
LOF	Leap of Faith Health Tech - the industry partner for CS595; operator of TheraCareAI
TheraCareAI	LOF's existing medication adherence tracking product; DischargeIQ is designed as the upstream comprehension layer that feeds into TheraCareAI's behavioral tracking pipeline

2g. Skills Needed

Team skillset required across 5 graduate students:

Skill Area	Team Members	Specific Tasks
Python backend (FastAPI, orchestration)	1-2 students	REST API design, async agent orchestration, inter-agent data passing, PARTIAL confidence warning implementation, error handling
LLM prompt engineering (Claude API)	1 student	System prompt design for each of five agents, chain-of-thought prompting, output validation
Frontend development (Streamlit or React)	1 student	Mobile-responsive UI, document upload interface, structured output rendering
Database and data modeling (Neon PostgreSQL)	1 student	Schema design for patient history, CRUD operations, discharge document storage
Evaluation and clinical validation	All team members (1 lead)	Flesch-Kincaid measurement, LLM-as-Judge rubric design, clinician template review coordination

3. System Overview

3a. Architecture

DischargeIQ is a five-agent sequential AI pipeline orchestrated by a FastAPI backend. Each agent handles a single, well-defined task type. A patient uploads a discharge PDF via a mobile-responsive web interface. The document passes through five agents in order. Explicit error handling at each handoff ensures a partial extraction still produces useful downstream output rather than failing silently.

Layer	Component	Role
Input	Mobile web UI (Streamlit or React)	Patient uploads discharge PDF; photo upload deferred to stretch goal
Orchestration	FastAPI backend	Manages sequential agent execution; explicit error states at each handoff; partial extraction produces partial output, never silent failure
Agent 1	Extraction Agent	Parses discharge doc to structured JSON; null for missing fields; never fabricates values
Agent 2	Diagnosis Explanation Agent	Plain-language condition explanation from one of 5 clinician-validated templates (AHA, GOLD, ADA, AAOS, ACC)
Agent 3	Medication Rationale Agent	Per-drug explanation; never advises stopping or changing medication; patient education only
Agent 4	Recovery Trajectory Agent	Week-by-week recovery guide with normal vs. alarming symptom distinctions for all 5 target diagnoses
Agent 5	Alert and Escalation Agent	Three-tier decision tree: normal / call doctor / go to ER; supplements document red flags with condition-specific flags
Output layer	Delivery layer (mobile + optional SMS)	Renders structured output; SMS via Twilio (stretch goal); not an agent
Storage	Neon PostgreSQL	Stores document hash, extracted fields, output versions per returning user
LLM backbone	Claude API (Anthropic)	Powers all five agents via specialized per-agent system prompts

3b. Technologies Used

Category	Technology	Purpose
Language	Python 3.11+	All backend development
Backend	FastAPI	REST API, agent orchestration, async handling, error state management between agents
LLM	Claude API (Anthropic)	All five agent tasks via specialized per-agent system prompts
Frontend	Streamlit or React	Mobile-responsive web UI; team decides in Sprint 1 Week 1

Category	Technology	Purpose
Database	Neon PostgreSQL	Patient history, extracted fields, output versioning
Readability	textstat (Python)	Flesch-Kincaid scoring; built into pipeline from Sprint 1 Week 1; used for prompt iteration throughout Sprint 2
Evaluation	LLM-as-Judge (Claude API)	Structured multi-dimensional scoring across 20 test cases; API costs estimated before Sprint 2
Notifications	Twilio (stretch goal)	SMS daily checklist; not required for MVP
Infrastructure	FastAPI + Redis + Neon	Open-source reference implementation from course
Version control	GitHub	Shared repository: one branch per agent recommended

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3c. Dependencies

Dependency	Type	Notes
Claude API (Anthropic)	External API	Primary LLM for all five agents; API key required; Claude API costs must be budgeted before Sprint 2 to cover LLM-as-Judge evaluation calls across 20 test cases
Neon PostgreSQL	Cloud database	Serverless Postgres; free tier sufficient for MVP; connection string distributed before Sprint 1
Twilio	External API	SMS checklist delivery; stretch goal only; not required for MVP
textstat	Python library	pip install textstat; local FK computation; no external calls
FastAPI + Uvicorn	Python library	pip install fastapi uvicorn; core backend runtime
Streamlit or React	Frontend	Team decides by Sprint 1 Week 1
pypdf / pdfplumber	Python library	PDF parsing; primary input format for MVP
Synthetic test documents	Data	10 de-identified or synthetic PDFs; must be ready before Sprint 1 Week 1 extraction testing begins
Clinical guidelines	Reference	AHA, GOLD, ADA, AAOS, ACC: one team member assigned per guideline for template validation; read-only reference

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3d. Hardware and Software Requirements

Development

Requirement	Specification
OS	macOS, Windows 10+, or Ubuntu 20.04+
Python	3.11 or higher
RAM	8 GB minimum; 16 GB recommended

Requirement	Specification
Node.js	18+ (only if React frontend is chosen)
IDE	VS Code recommended
API credentials	Anthropic API key and Neon PostgreSQL connection string; distributed before Sprint 1 starts
Internet	Required for Claude API calls and Neon database access

Testing and Deployment

All testing runs locally or against a staging environment. Test suite: pytest for backend unit tests, Chrome mobile emulation for the frontend, and 10 synthetic discharge documents as the primary test corpus. Per the professor's recommendation, the Extraction Agent must be tested against all 10 documents in Sprint 1 Week 1 and a baseline documented before any downstream agents are built.

The MVP prototype runs locally or on a free-tier cloud host (Render, Railway, or Streamlit Cloud). No hospital infrastructure integration required. HIPAA-compliant infrastructure is excluded from MVP and documented as a Phase 2 requirement.

3e. Data Description and Sources

Data Type	Source	Size and Accessibility
Synthetic discharge documents	Team-generated for 5 target diagnoses	10 PDFs in shared GitHub repo under /test-data; required before Sprint 1 Week 1
De-identified public examples	CMS public use files, MITRE synthetic health data	5 to 10 docs; publicly available
Clinical guideline references	AHA, GOLD (COPD), ADA (diabetes), AAOS (orthopedic), ACC (surgical)	Read-only web references; one team member assigned per guideline
Drug reference data	FDA drug label database (DailyMed)	Web reference only; no local copy required for MVP
Patient history (production)	User-uploaded discharge documents; document hash and structured fields only in Neon PostgreSQL	Per-user; full document text not stored; encrypted at rest

The MVP operates exclusively on de-identified and synthetic documents. No real patient health information enters the system during the academic build.

3f. Pre-Work and Setup

Task	Owner	When
Create shared GitHub repository; agree on one-branch-per-agent strategy	Team lead	Before Sprint 1 Week 1

Task	Owner	When
Obtain and distribute Anthropic API keys	Team lead	Before Sprint 1 Week 1
Provision Neon PostgreSQL instance; share connection string	Backend owner	Before Sprint 1 Week 1
Decide Streamlit vs React for frontend	Full team	Sprint 1 Week 1
Collect 10 synthetic or de-identified discharge PDFs covering all 5 diagnoses	All members	Sprint 1 Week 1
Define and document Extraction Agent JSON schema before any code is written	Backend owner	Sprint 1 Week 1
Set up local Python environments; install base dependencies	All members	Sprint 1 Week 1
Assign one team member to each of the 5 clinical guidelines for template validation	Team lead	Sprint 1 Week 1
Estimate Claude API cost for 20-case LLM-as-Judge evaluation; set budget ceiling	Backend owner	Before Sprint 2 starts

4. Functional Requirements

4a. List of User Tasks

Tasks listed in Verb + Noun format for use in usability testing.

#	User Task	Actor	Sprint
1	Upload discharge document	Patient or Caregiver	Sprint 1
2	View extracted diagnosis summary	Patient or Caregiver	Sprint 2
3	Read medication explanations	Patient or Caregiver	Sprint 2
4	View recovery timeline	Patient or Caregiver	Sprint 2
5	Check escalation decision tree	Patient or Caregiver	Sprint 2
6	View past discharge summaries	Returning user	Sprint 2
7	Opt in to SMS checklist reminders	Patient or Caregiver	Stretch

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4b. User Stories

US-01: As a recently discharged patient, I want to upload a PDF of my discharge document so that I can get a plain-language explanation of what happened to me and what I need to do next.
US-02: As a patient with low health literacy, I want my medications explained in simple language that connects each drug to my diagnosis so that I understand why I am taking each one.
US-03: As a caregiver managing a family member's recovery, I want a clear recovery timeline so that I know what is normal and what should concern me in the first two weeks.
US-04: As a patient recovering at home at 2am, I want a clear escalation guide that tells me exactly which symptoms mean call the doctor and which mean go to the ER so that I do not have to guess.
US-05: As a patient who did not fully understand the nurse's verbal instructions, I want a persistent written reference I can return to at any time so that critical information is not lost.
US-06: As a returning user with a chronic condition, I want to view my past discharge summaries so that I can track changes across hospitalizations.
US-07: As a hospital quality improvement director, I want patients who use DischargeIQ to show measurable improvement in comprehension scores so that I can justify deploying it as a discharge intervention.

4c. Use Cases

Use Case	Actor	Precondition	Steps	Outcome
UC-01: Upload and extract	Patient	Has a discharge PDF	1. Open DischargeIQ 2. Tap upload 3. Select PDF 4. Confirm	Extraction Agent returns structured JSON; patient sees extracted fields

Use Case	Actor	Precondition	Steps	Outcome
UC-02: Read output	Patient	Extraction successful	1. View five-section output 2. Scroll sections 3. Tap for detail	Patient reads output at 6th grade reading level per FK scorer
UC-03: Escalation check	Patient at 2am	Experiencing a symptom	1. Open escalation section 2. Find symptom 3. Read action	Unambiguous call doctor / go to ER / normal instruction
UC-04: History review	Returning patient	Prior upload in database	1. Log in 2. Select past discharge 3. View output	Prior output retrieved from Neon without re-processing

4d. Prioritization

Requirements prioritized using MoSCoW: Must have, Should have, Could have, Will not have for this MVP.

Requirement	Priority	Sprint
Discharge document upload (PDF)	Must have	Sprint 1
Extraction Agent with validated JSON schema output	Must have	Sprint 1
Explicit error handling at each agent handoff	Must have	Sprint 1
FK scorer integrated into pipeline from Sprint 1	Must have	Sprint 1
Diagnosis Explanation Agent (5 templates, clinician-reviewed)	Must have	Sprint 2
Medication Rationale Agent	Must have	Sprint 2
Recovery Trajectory Agent	Must have	Sprint 2
Alert and Escalation Agent	Must have	Sprint 2
Mobile-responsive frontend (Streamlit or React)	Must have	Sprint 2
Patient history storage (Neon PostgreSQL)	Must have	Sprint 2
Evaluation framework: 20 test cases, LLM-as-Judge rubric	Must have	Sprint 2
Photo and camera upload with OCR	Should have	Stretch
SMS daily checklist via Twilio	Could have	Stretch
Physician read-only summary view	Could have	Stretch
Diagnosis templates beyond core 5	Could have	Stretch
HIPAA-compliant infrastructure	Will not have (MVP)	Phase 2
EHR integration (Epic, Cerner)	Will not have (MVP)	Phase 2

Requirement	Priority	Sprint
Clinical decision support or differential diagnosis	Will not have (ever)	Out of scope

4e. List of Functional Requirements

FR #	Requirement	Agent or Component
FR-01	System shall accept PDF discharge document uploads via mobile web interface; photo upload is a stretch goal	Frontend + FastAPI
FR-02	Extraction Agent shall return a valid JSON object matching the defined output schema for every input	Agent 1
FR-03	Extraction Agent shall return null for fields it cannot extract; it shall never fabricate a value	Agent 1
FR-04	FastAPI orchestration layer shall pass structured output to the next agent even on partial extraction, with explicit error states surfaced to the user rather than silent failure	FastAPI
FR-05	Diagnosis Explanation Agent shall generate a plain-language explanation from one of the 5 clinician-validated templates, reviewed against its assigned clinical guideline	Agent 2
FR-06	All agent text outputs shall be scored by the FK scorer; outputs exceeding 6th grade shall be flagged for prompt revision	FK scorer
FR-07	Medication Rationale Agent shall connect each medication to the patient's specific diagnosis in plain language	Agent 3
FR-08	Medication Rationale Agent shall never advise stopping or changing a medication; all output must remain patient education	Agent 3
FR-09	Recovery Trajectory Agent shall generate a week-by-week recovery guide with explicit normal vs. alarming symptom distinctions	Agent 4
FR-10	Alert and Escalation Agent shall produce a three-tier decision tree (normal / call doctor / go to ER) for every output	Agent 5
FR-11	Alert and Escalation Agent shall supplement document red flags with condition-specific flags not always listed on discharge paperwork	Agent 5
FR-12	System shall store document hash, extracted fields, and output version in Neon PostgreSQL for returning users	Storage
FR-13	Frontend shall display output in five clearly labeled sections readable on a smartphone without horizontal scrolling	Frontend
FR-14	System shall display a plain-language disclaimer that output is patient education only and does not replace clinical advice	Frontend

4f. Data Structures and Interfaces

Extraction Agent Output Schema

The following schema is the contract between Agent 1 and all downstream agents. It must be defined and documented in the repository before any agent code is written. This is a Sprint 1 Week 1 prerequisite.

{ "patient_name": "string | null", "discharge_date": "ISO date | null", "primary_diagnosis": "string", "secondary_diagnoses": ["string"], "procedures_performed": ["string"], "medications": [{ "name": "string", "dose": "string | null", "frequency": "string | null", "duration": "string | null", "status": "new | changed | continued | discontinued" }], "follow_up_appointments": [{ "provider": "string | null", "specialty": "string | null", "date": "string | null", "reason": "string | null" }], "activity_restrictions": ["string"], "dietary_restrictions": ["string"], "red_flag_symptoms": ["string"], "discharge_condition": "string | null" }

Agent-to-Agent Interface

Each agent receives the full accumulated context object from the FastAPI orchestration layer. Error states from any agent are included so downstream agents can produce partial output rather than failing.

Agent	Input	Output added to context
Agent 1: Extraction	Raw document (PDF bytes)	Full structured JSON per schema; error state included if fields missing
Agent 2: Diagnosis Explanation	Primary and secondary diagnoses from Agent 1	Plain-language condition explanation from validated template
Agent 3: Medication Rationale	Medications list + primary diagnosis context	Per-medication explanation objects with side effects and dose-miss guidance
Agent 4: Recovery Trajectory	Primary diagnosis + procedures performed	Week-by-week recovery guide with normal vs. alarming symptom distinctions
Agent 5: Alert and Escalation	Red flag symptoms + primary diagnosis context	Three-tier escalation object: normal / call doctor / go to ER

4g. User Tasks: Usability Testing Format

Task #	Task Instruction	Success Criterion
T-01	You have just been discharged from the hospital and have your discharge papers. Use DischargeIQ to upload them and find out what your main diagnosis means.	User completes upload and reaches diagnosis explanation section without assistance
T-02	You were prescribed three new medications. Use DischargeIQ to find out what each one is for and why you are taking it.	User locates medication section and reads all three explanations without confusion
T-03	It is Day 5 after your discharge. You feel a sharp pain in your chest. Use DischargeIQ to decide what to do.	User reaches escalation section and identifies correct action tier without assistance

Task #	Task Instruction	Success Criterion
T-04	You want to know whether you are allowed to climb stairs yet. Use DischargeIQ to find out.	User locates recovery trajectory section and receives a clear answer about activity restrictions
T-05	You were discharged two months ago and want to look back at what your recovery plan said. Find your previous discharge summary.	User locates history section and retrieves prior output without re-uploading

5. General Schedule

DischargeIQ will be built over two sprints in a six-week timeline. The professor's primary instruction: build agents incrementally. Confirm Agent 1 output matches the schema before writing Agent 2. Confirm Agents 1 and 2 together before adding Agent 3. Continue this pattern through Agent 5. Do not attempt to integrate all five agents simultaneously. The FK scorer is built in Sprint 1 so prompt iteration throughout Sprint 2 uses real reading level data.

Sprint 1: March 24 to April 6, 2026

Sprint 1 delivers the backend foundation, Agents 1 through 3, the FK scorer, and 3 of the 5 diagnosis templates drafted and internally reviewed. Agents are built incrementally: Agent 2 is started only after Agent 1 is confirmed, and Agent 3 is started only after Agents 1 and 2 are confirmed together end-to-end.

Week 1: Foundation, Extraction Baseline and FK Integration

Task	Detail
Set up FastAPI backend, Neon PostgreSQL, and shared repo	Shared GitHub repo with one branch per agent; Neon PostgreSQL schema initialized; Claude API connected and verified with a test call
Write and lock the Agent 1 JSON output schema	Fields: patient name, discharge date, primary and secondary diagnoses, medications (name/dose/frequency/duration/status), follow-up appointments, activity restrictions, dietary restrictions, red flag symptoms, discharge condition. Missing fields return null, never fabricated. Schema documented in repo before any code is written.
Collect 10 synthetic or de-identified discharge PDFs	Covering all 5 target diagnoses: heart failure, COPD, diabetes management, hip replacement, one surgical case. In shared repo under /test-data by end of Week 1.
Integrate FK scorer into the pipeline this week	textstat library, two lines of code. Runs on every Agent 1 output from day one so real FK data is available for prompt iteration throughout Sprint 1 and Sprint 2.
Build and test Agent 1 against all 10 documents	Do not move to Agent 2 until valid JSON is returned on at least 8 of 10 with no fabricated values. Log extraction accuracy per document. Extraction baseline documented.
Assign one team member per clinical guideline	AHA (heart failure), GOLD (COPD), ADA (diabetes management), AAOS (hip replacement), ACC (surgical case). Confirmed by end of Week 1 so template drafting in Week 2 has clear ownership.

Task	Detail
Estimate Claude API costs for LLM-as-Judge evaluation	Estimate cost for 20-case batch before Sprint 2 begins. Budget ceiling agreed with the team.

Week 2: Agents 2 and 3 + Template Drafting

Task	Detail
Build Diagnosis Explanation Agent (Agent 2)	On confirmed Agent 1 output only. FK score confirmed at or below 6th grade on all outputs before Agent 3 is started.
Build Medication Rationale Agent (Agent 3)	On confirmed Agent 1 plus 2 output. Never advises stopping or changing a medication. Output stays strictly within patient education framing.
Draft 3 of the 5 diagnosis templates	Heart failure, COPD, and diabetes management. Each reviewed against the assigned clinical guideline by the responsible team member before being used in the pipeline.
Sprint 1 demo	Upload a PDF, show Agents 1 through 3 running end-to-end, show FK scores in the pipeline log. Ensure realistic test output is ready before the demo date, not just a pipeline that technically runs.

Sprint 1 Exit Criteria

Criterion	Standard
Agents 1 through 3 running sequentially with confirmed handoffs	Each handoff tested end-to-end on multiple documents, not just a single run
Valid output on at least one full test document end-to-end	Realistic output, not placeholder text
FK scorer live in the pipeline from Week 1	Scoring every agent output and logging results
3 diagnosis templates drafted and internally reviewed against assigned clinical guidelines	Heart failure, COPD, diabetes management; reviewed by assigned team member against AHA, GOLD, ADA
API cost estimate for LLM-as-Judge evaluation documented	Budget ceiling agreed before Sprint 2 begins

Sprint 2: April 7 to May 4, 2026

Sprint 2 runs from April 7 to May 4, 2026 across four weeks. Week 3 builds Agents 4 and 5 individually with confirmed handoffs. Week 4 handles full integration and error handling. Week 5 runs evaluation. Week 6 polishes and delivers the final demo.

Week 3: Agents 4 and 5

Task	Detail
Build Recovery Trajectory Agent (Agent 4)	On confirmed Agents 1 through 3 output only. Do not wire into the full pipeline until Agent 4 output is confirmed stable on multiple test documents.
Build Alert and Escalation Agent (Agent 5)	On confirmed Agents 1 through 4 output only. Confirm all 5 handoffs before wiring through FastAPI.

Week 4: Full Integration and Error Handling

Task	Detail
Wire all 5 agents through FastAPI + Redis	Once Agents 4 and 5 are individually confirmed. Full pipeline runs end-to-end on all 10 test documents.
PARTIAL or LOW confidence warning banner	PARTIAL or LOW confidence from Agent 1 must surface a visible warning banner, never a crash. Downstream agents pass extraction_warnings[] and still produce best-effort output.
Pipeline performance	Time the pipeline on all 10 documents. Target P95 under 30 seconds. Record actual numbers.
Mobile frontend test	Test on a real phone. If output scrolls sideways or needs pinch-to-zoom, fix it before Week 5.
Neon PostgreSQL session history	Returning user retrieves prior output without re-uploading. Document hash and structured fields only, no PHI stored.
Bug logging and P0 fixes	Log every bug found with P0/P1/P2 severity. All P0s fixed before Week 5 starts.
Get faculty sign-off on all 5 templates with guideline citations	ACC/AHA (heart failure), GOLD 2024 (COPD), ADA 2024 (diabetes management), AAOS (hip replacement), ACC/ACS perioperative guidelines (surgical case).

Week 5: Evaluation

Task	Detail
Run LLM-as-Judge batch on 20 test cases	4 per diagnosis. Score: Clinical Accuracy, Plain Language, Completeness, Actionability (each 1 to 5) and Safety (Pass/Fail). Roughly 100 API calls, around $3 to $8. Run once as a batch using the judge prompt drafted in Sprint 1. Any Safety fail is a P0 fixed before the final demo. Target average Clinical Accuracy at or above 3.5.
Calculate FK delta for all 10 documents	DischargeIQ output must score lower than the raw discharge document on every one. Report delta per diagnosis.
Format evaluation results into a pre-prepared summary	FK delta table, judge scores, failure mode log. Finished before the demo, not calculated live during the demo.
Clear all P1 bugs and clean up the frontend	Demo flow must run clean: upload, five sections, done.

Week 6: Polish and Delivery

Task	Detail
Finalize evaluation report and pilot deployment doc	Evaluation report with FK delta table, judge scores, and failure modes. Pilot deployment doc covering data handling scope and HIPAA Phase 2 requirements. Both go in the presentation appendix.
Full dry run of the final demo	Upload a PDF, walk all 5 sections, show FK scores live, present the pre-prepared evaluation summary.

Task	Detail
Stretch goals	Only if the pipeline is fully stable on Day 1 of Week 6. One goal maximum, secondary owner only: photo OCR via Claude Vision, SMS checklist via Twilio, or physician summary view.

Sprint 2 Exit Criteria

Criterion	Standard
All 5 agents connected	10/10 test documents produce output with no silent failures
PARTIAL confidence warning banner live and tested	Visible warning shown on partial extractions; no crashes
P95 pipeline time under 30 seconds	Documented with actual numbers from all 10 test documents
LLM-as-Judge complete	20 cases scored, zero Safety failures, average Clinical Accuracy at or above 3.5
FK delta positive for all 10 documents	Reported per diagnosis
All 5 templates with faculty sign-off on file	With guideline citations attached
Mobile frontend tested on 2 devices	No scroll or zoom issues
Neon PostgreSQL session history working for returning users	Document hash and structured fields only, no PHI
Evaluation summary pre-prepared and ready for demo	Not calculated live
Final demo dry run done and passing	Full pipeline on a real or emulated mobile device

Submission of Results

#	Deliverable	Format and Location
1	All source code for the five-agent pipeline, FastAPI backend, and frontend	GitHub repository link submitted to course platform
2	All AI prompts: system prompts for each of the five agents and the LLM-as-Judge evaluation rubric, versioned with iteration notes	Repo under /prompts
3	Synthetic and de-identified test documents used for development and evaluation	Repo under /test-data; all confirmed de-identified
4	Evaluation results: extraction accuracy logs, FK scores per output, LLM-as-Judge results for 20 test cases, API cost actuals	Repo under /evaluation; summary table in final report
5	Clinician-reviewed explanation templates for 5 target diagnoses with validation notes and guideline citations	Repo under /templates

#	Deliverable	Format and Location
6	Final project report covering architecture, evaluation results, known limitations, and Phase 2 roadmap	PDF submitted to course platform
7	Live demo or recorded walkthrough of the working system	Demonstrated in class or submitted as video link
8	Data pipeline documentation: schema definitions, agent interfaces, error state handling	Markdown files in repo under /docs

6. Addendum: Pillar Alignment and Sprint Execution Plan

6.1 Primary Pillar Alignment

Patient Engagement

6.2 LOF Healthcare Pillar Justification

DischargeIQ's primary users are patients and caregivers recently discharged from acute care hospitals in the US who need to independently understand and act on their discharge documents at home. The healthcare problem it addresses is the discharge comprehension gap: 40-80% of patients forget or misunderstand critical discharge instructions, patients who make comprehension errors are nearly 9 times more likely to make adherence errors, and no existing tool connects a patient's specific diagnosis, medications, and recovery trajectory in a single plain-language output. DischargeIQ creates value under the Patient Engagement pillar by transforming complex clinical discharge documents into structured, verified, plain-language guidance at a 6th grade reading level that patients can access on any mobile device, reducing the comprehension failure that contributes to $41.3 billion in annual readmission costs, 27% of which is estimated preventable. Every design decision in the system, from the 6th grade reading level target to the three-tier escalation guide, is oriented around activating the patient as an informed participant in their own recovery rather than a passive recipient of a clinical document.

6.3 Evidence from the Proposal

- Patients who make comprehension errors are nearly 9 times more likely to make adherence errors (Section 2a Need). This statistic establishes that patient comprehension is a clinical safety issue, which is precisely the gap the Patient Engagement pillar targets.
- The system is designed for a 6th grade reading level with Flesch-Kincaid scoring built into the pipeline from Sprint 1 and applied to every agent output. This is an enforced design constraint and the defining characteristic of a patient-engagement tool versus a clinical documentation tool.
- The evaluation framework measures pre/post comprehension scores as the primary success metric, not provider satisfaction or clinical workflow integration, confirming that patient understanding and behavior change are the project's core outcome measures.

6.4 Sprint Execution Plan

Sprint 1: March 24 to April 6, 2026

Week 1 establishes the full foundation: FastAPI backend, Neon PostgreSQL, shared repo, Agent 1 JSON schema locked before any code is written, 10 synthetic discharge PDFs collected, FK scorer integrated from day one, Agent 1 built and tested with a hard gate of valid JSON on at least 8 of 10 documents before proceeding, one team member assigned per clinical guideline, and Claude API cost estimate completed. Week 2 builds Agent 2 (Diagnosis Explanation) on confirmed Agent 1 output, then Agent 3 (Medication Rationale) on confirmed Agents 1 and 2 output, with 3 of the 5 diagnosis templates drafted and internally reviewed against assigned guidelines. Sprint 1 demo shows Agents 1 through 3 running end-to-end with FK scores in the pipeline log.

Sprint 2

Week 3 builds Agents 4 (Recovery Trajectory) and 5 (Alert and Escalation) individually on confirmed prior output. Week 4 wires all 5 agents through FastAPI and Redis, implements the PARTIAL confidence warning banner, times

the pipeline on all 10 documents, gets Neon PostgreSQL session history working, and obtains faculty sign-off on all 5 templates. Week 5 runs the LLM-as-Judge batch evaluation on 20 test cases, calculates FK delta, and prepares the evaluation summary. Week 6 finalizes the evaluation report and pilot deployment doc, does a full dry run of the final demo, and attempts stretch goals only if the pipeline is fully stable on Day 1 of Week 6, one goal maximum, secondary owner only.

6.5 Scope Confirmation

Plan Status: This sprint plan represents a refinement and strategic narrowing of our original scope.

Refinement Justification

- Focusing on High-Impact Diagnoses: Based on instructor feedback, we narrowed the MVP from ten diagnosis templates to five core pathways (Heart Failure, COPD, Diabetes Management, Hip Replacement, and one surgical case). This ensures each pathway is clinically accurate and deeply validated rather than superficially covering too many conditions.
- Prioritizing Patient Safety: By concentrating our prompt engineering on these five core areas, we can guarantee that all AI-generated instructions strictly adhere to the 6th-grade Flesch-Kincaid reading level required for safe patient comprehension at home.
- Managing Technical Complexity: Our five-agent architecture is powerful but intricate. Narrowing the diagnosis count allows the team to dedicate more effort to building a Robust Error Handling Layer and a Partial Confidence Warning Banner, ensuring any extraction issues from Agent 1 are caught and flagged before affecting downstream agents.
- Rigorous Testing and Evaluation: We shifted resources from content volume to quality verification. Using an LLM-as-Judge framework, we will score 20 test cases on Clinical Accuracy, Actionability, and Safety, ensuring the pipeline is stable and reliable for our final demo.
- Defining MVP Boundaries: This refined scope keeps the project focused on Patient Engagement. It allows us to deliver a polished, high-quality tool that solves the discharge comprehension gap while avoiding the technical debt or safety risks of an over-ambitious initial scope.

6.6 Healthcare Data, Standards, and Clinical Context

DischargeIQ does not require direct integration with a formal data standard for the MVP but is grounded in real clinical workflows and published clinical evidence throughout. The five-agent pipeline is structured around the actual content of hospital discharge documents as produced by Epic and equivalent EHR systems. Clinician-validated explanation templates are developed against published clinical guidelines for the 5 target diagnoses: AHA for heart failure, GOLD for COPD, ADA for diabetes management, AAOS for hip replacement, and ACC for the surgical case. Reading level is measured using Flesch-Kincaid scoring against the AMA-recommended 6th grade target. The exclusion of clinical decision support is a deliberate regulatory decision keeping the system within patient education scope and outside FDA oversight. The prototype operates exclusively on de-identified and synthetic discharge documents. HIPAA-compliant infrastructure is a Phase 2 requirement.